

Congruence and Primes

- (Chinese remainder theorem) Let $n_1, n_2, \dots, n_k$ be pairwise relatively prime positive integers, and let $a_1, a_2, \dots, a_k$ be integers. Then the following system of congruence equations has a unique solution modulo $n_1 n_2 \cdots n_k$.

$$\begin{cases} m \equiv a_1 \pmod{n_1} \\ m \equiv a_2 \pmod{n_2} \\ \dots \\ m \equiv a_k \pmod{n_k} \end{cases}$$

- (Euler-Fermat theorem) Let n be a positive integer and a an integer with $(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

- (Fermat little theorem) Let p be a prime number and a an integer with $(a, p) = 1$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

- (Wilson's theorem) Let p be a prime number. Then $(p-1)! \equiv -1 \pmod{p}$.
- (Primitive root) Let p be an odd prime, and let k be a positive integer. Then there exists an integer g such that $g, g^2, \dots, g^{\varphi(p^k)}$ are pairwise incongruent modulo p^k and they run through all congruence classes modulo p^k relatively prime to p .
- (Lifting the exponent lemma) Let p be an odd prime, n a positive integer and a, b distinct integers with $a \equiv b \not\equiv 0 \pmod{p}$. Then $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$.
- (Zsigmondy's theorem) Let $a > b \geq 1$ be relatively prime integers and n a positive integer. Then there exists a prime p such that $p \mid a^n - b^n$ and $p \nmid a^k - b^k$ for each $k = 1, 2, \dots, n-1$ except the following cases:

(i) $a - b = 1$ and $n = 1$;

(ii) $a + b$ is a power of 2 and $n = 2$;

(iii) $a = 2, b = 1$ and $n = 6$.

- (Dirichlet's theorem) Let a and b be relatively prime integers where $a > 0$. Then there are infinitely many primes of the form $ak + b$.
- (Kobayashi's theorem) Let $a_1, a_2, \dots$ be an infinite sequence of distinct positive integers. If there are only finitely many primes p such that $p \mid a_k$ for some k , then for any positive integer m , there are infinitely many primes p such that $p \mid a_k + m$ for some k .
- (Cauchy-Davenport theorem) Let p be a prime number and let $A, B \subset \mathbb{F}_p$. Then

$$|A + B| \geq \min \{p, |A| + |B| - 1\}.$$

Problem 1. Let p be an odd prime. Let $a_1, a_2, \dots, a_p$ be a permutation of $1, 2, \dots, p$. Prove that there exist distinct integers $m, n \in \{1, 2, \dots, p\}$ such that $p \mid ma_m - na_n$.

Problem 2. Let p be a prime number such that $p \equiv 2 \pmod{3}$. Find

$$\prod_{k=1}^{p-1} (k^2 + k + 1) \pmod{p}.$$

Problem 3. Let n be a fixed positive integer. Consider a natural number $a > 1$ and different odd prime numbers $p_1, p_2, \dots, p_n$ with

$$a^{p_1} \equiv 1 \pmod{p_2}, \quad a^{p_2} \equiv 1 \pmod{p_3}, \quad \dots, \quad a^{p_n} \equiv 1 \pmod{p_1}.$$

- (a) Prove that $p_i \mid (a - 1)$ for some $i = 1, 2, \dots, n$.
- (b) Can $(a - 1)$ be divisible by p_i for exactly one $i \in \{1, 2, \dots, n\}$?

Problem 4. Let a and b be distinct positive integers. Prove that there are infinitely many prime numbers p such that

$$p \mid a^n + b^{n+1}$$

for some positive integer n .

Problem 5. Let p be a prime. Find all integers n such that for all integers x , if $x^n - 1$ is divisible by p , then $x^n - 1$ is divisible by p^2 .

Problem 6. Let $p \geq 7$ be a prime number. Prove that one of the following numbers is in form of $x^2 + y^2$.

$$p + 1, 2p + 1, 3p + 1, \dots, (p - 3)p + 1$$

Problem 7. Prove that for any given positive integers m, n , there exist infinitely many pairs of relatively prime positive integers a, b such that

$$a + b \mid am^a + bn^b.$$

Problem 8. Prove that for any prime p , there exists a positive integer n such that

$$1^n + 2^{n-1} + 3^{n-2} + \dots + n^1 \equiv 2020 \pmod{p}.$$